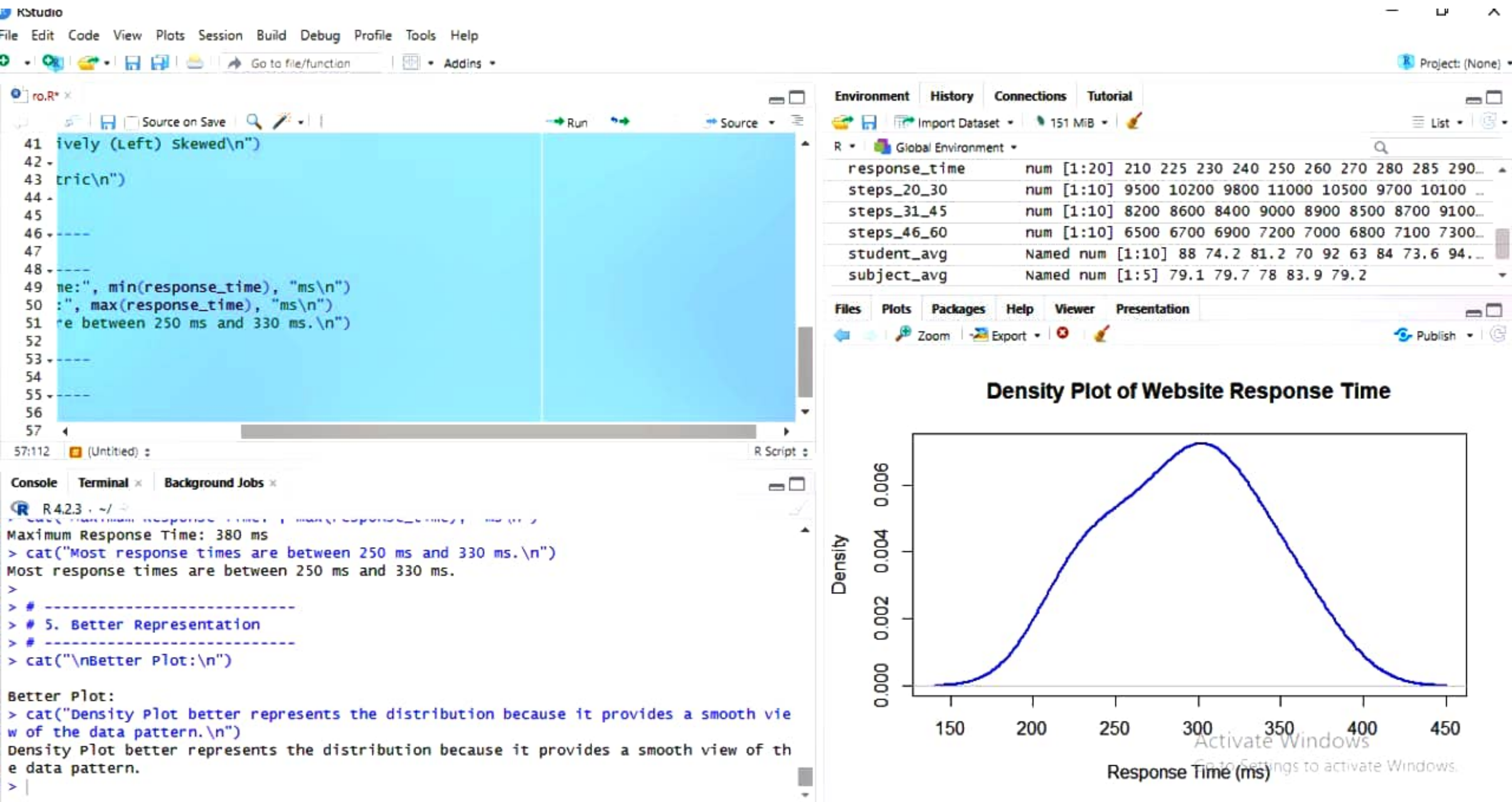
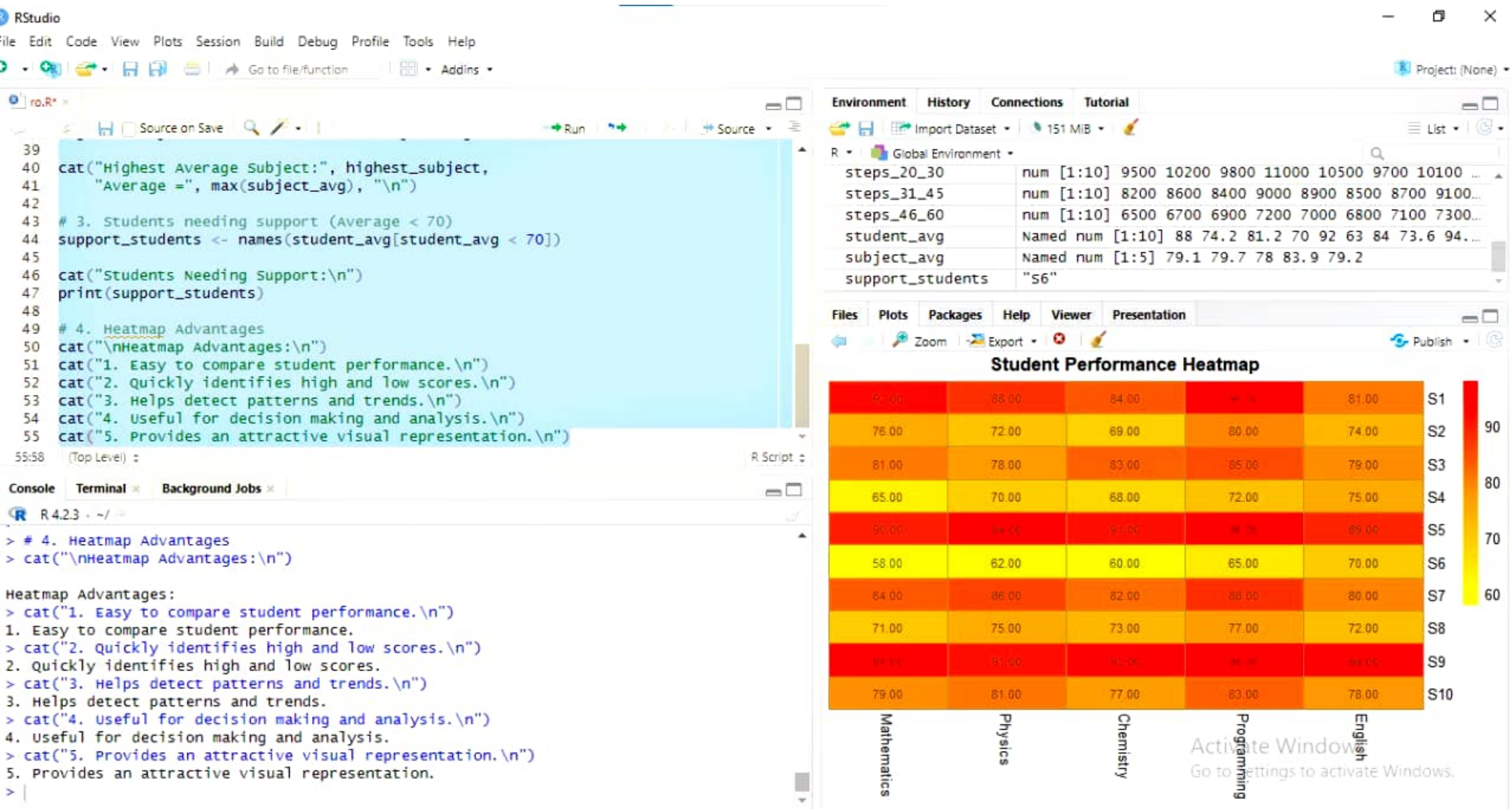
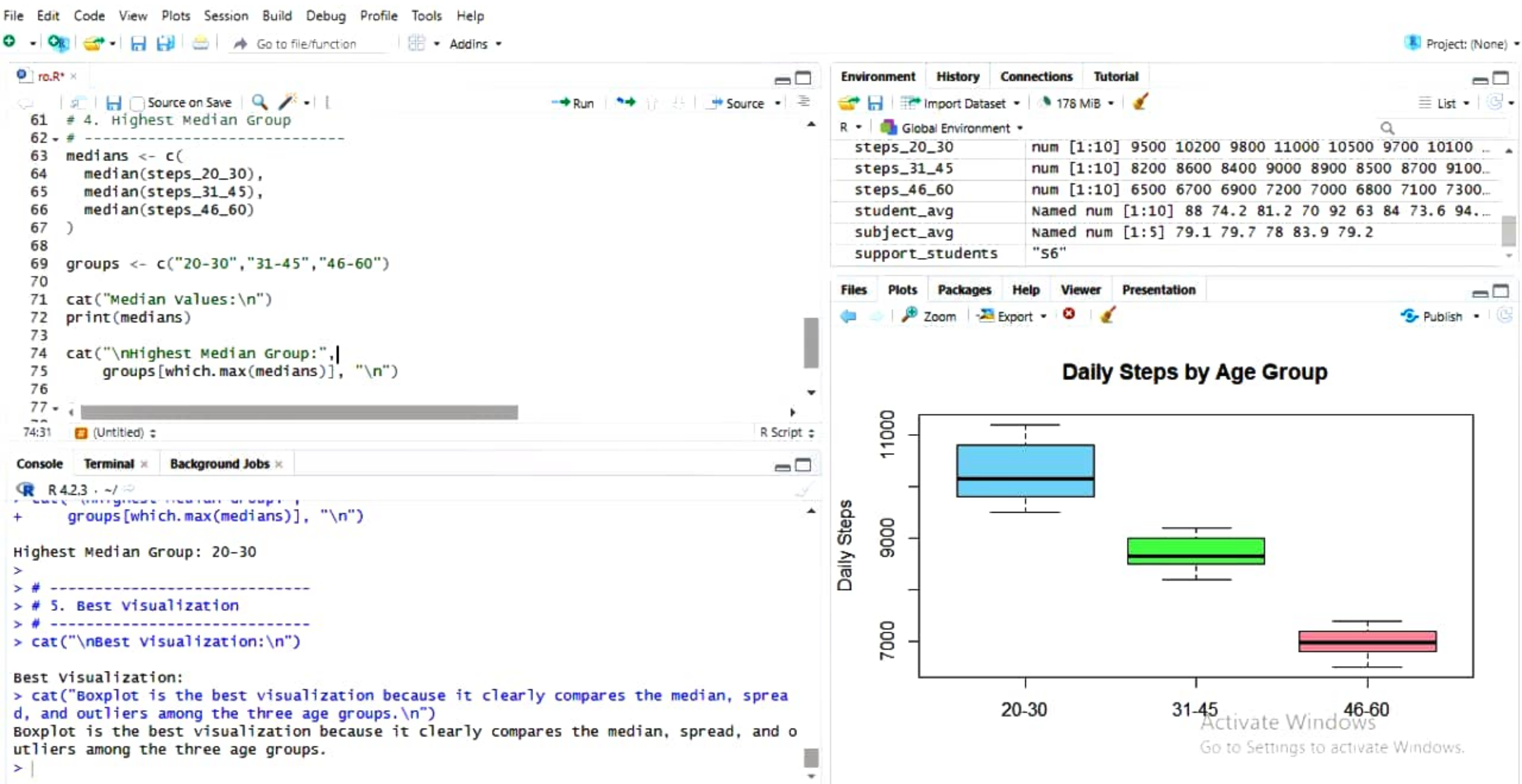
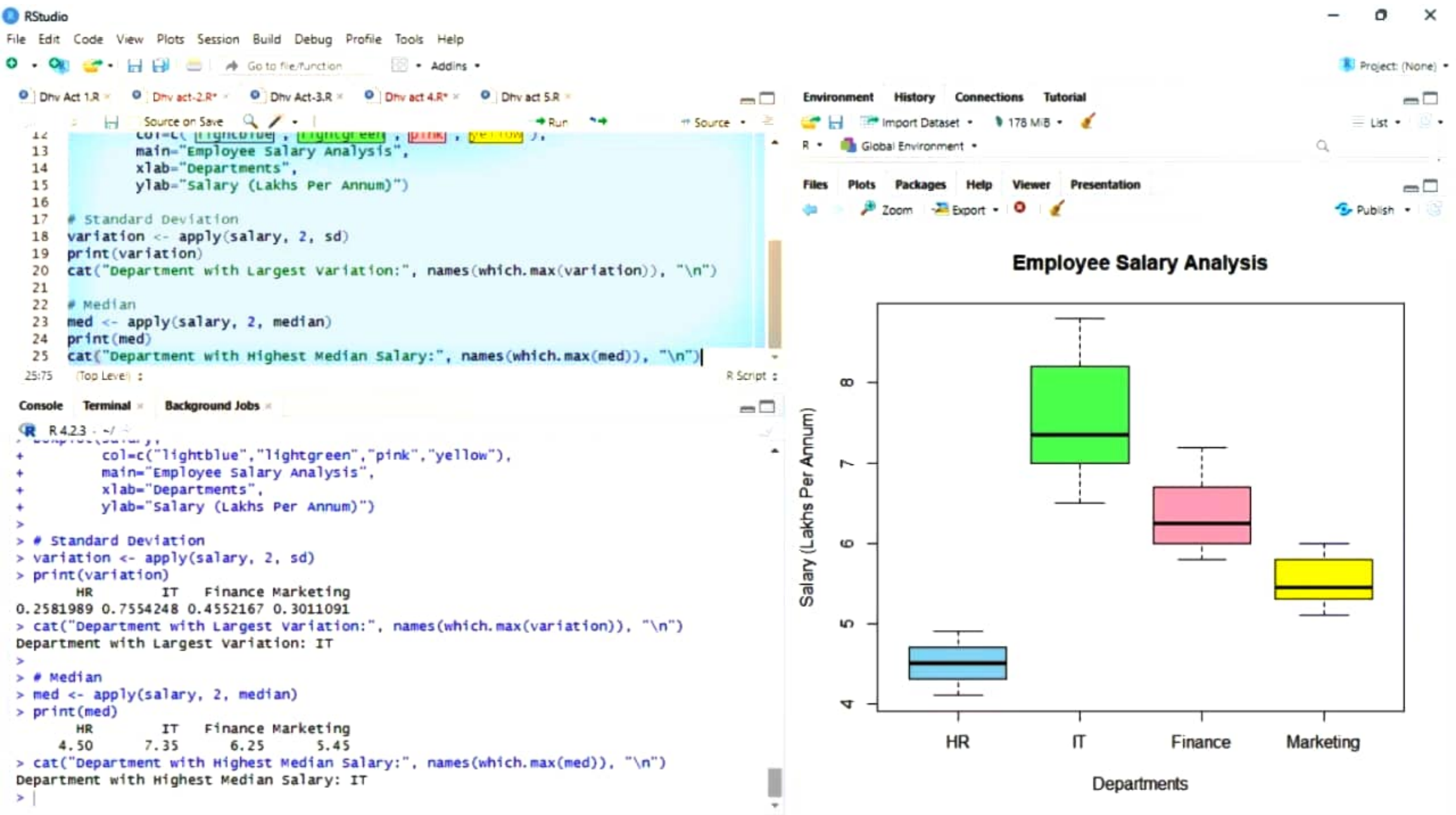
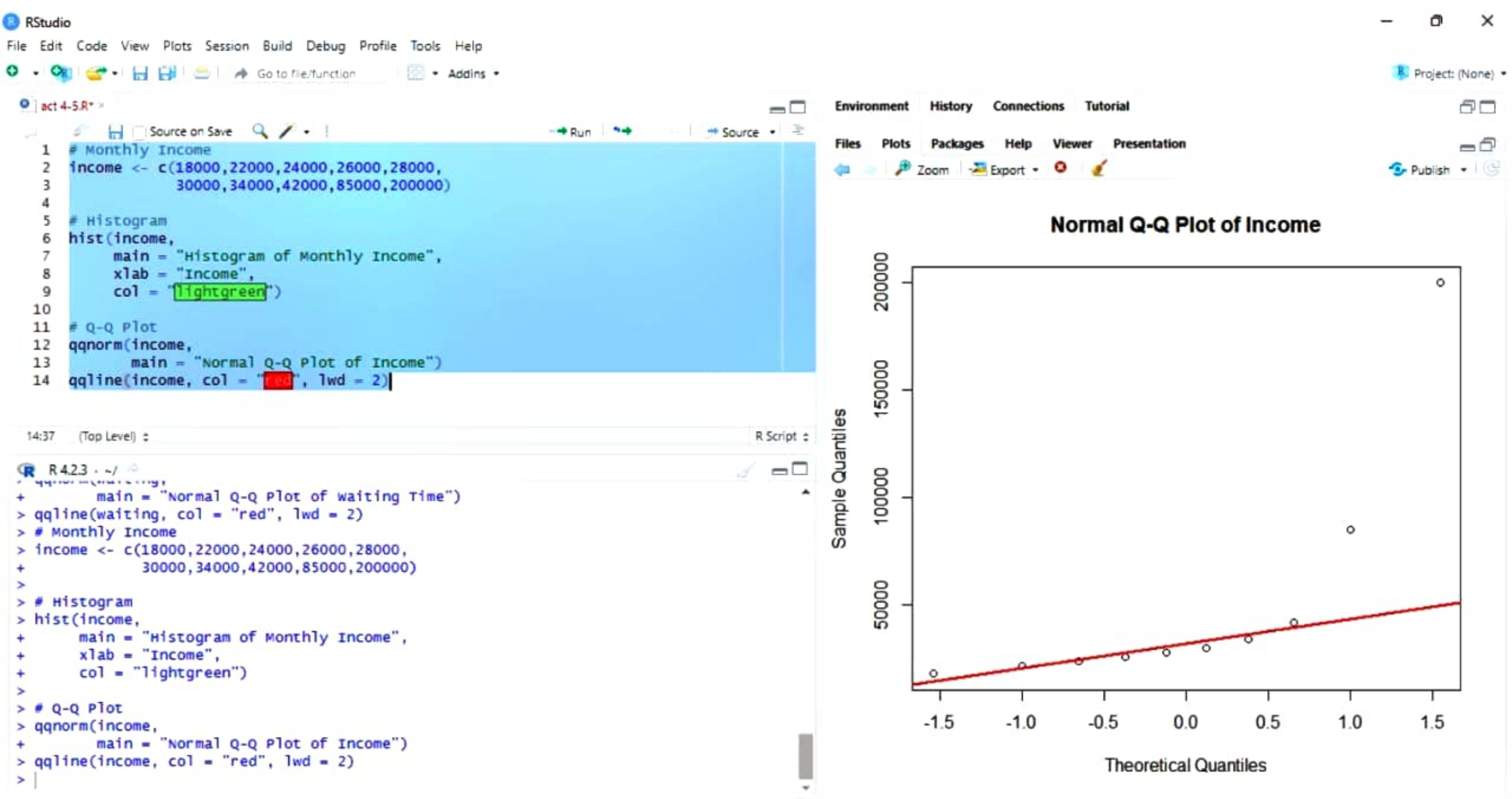

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
ro.R*
Source on Save
Run
Source
41 ively (Left) Skewed\n")
42
43 tric\n")
44
45
46 ----
47
48 ----
49 me:", min(response_time), "ms\n")
50 :", max(response_time), "ms\n")
51 re between 250 ms and 330 ms.\n")
52
53 ----
54
55 ----
56
57
57:112
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.2.3 · ~/
Maximum Response Time: 380 ms
> cat("Most response times are between 250 ms and 330 ms.\n")
Most response times are between 250 ms and 330 ms.
>
> # ---------------------------
> # 5. Better Representation
> # ---------------------------
> cat("\nBetter Plot:\n")
Better Plot:
> cat("Density Plot better represents the distribution because it provides a smooth vie
w of the data pattern.\n")
Density Plot better represents the distribution because it provides a smooth view of th
e data pattern.
>
Environment
History
Connections
Tutorial
Import Dataset
151 MiB
List
R
Global Environment
response_time num [1:20] 210 225 230 240 250 260 270 280 285 290…
steps_20_30 num [1:10] 9500 10200 9800 11000 10500 9700 10100 …
steps_31_45 num [1:10] 8200 8600 8400 9000 8900 8500 8700 9100…
steps_46_60 num [1:10] 6500 6700 6900 7200 7000 6800 7100 7300…
student_avg Named num [1:10] 88 74.2 81.2 70 92 63 84 73.6 94.…
subject_avg Named num [1:5] 79.1 79.7 78 83.9 79.2
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Density Plot of Website Response Time
Density
0.000
0.002
0.004
0.006
150
200
250
300
350
400
450
Response Time (ms)
Activate Windows
Go to Settings to activate Windows.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
ro.R*
Source on Save
Run
Source
cat("Highest Average Subject:", highest_subject,
"Average =", max(subject_avg), "\n")
# 3. Students needing support (Average < 70)
support_students <- names(student_avg[student_avg < 70])
cat("Students Needing Support:\n")
print(support_students)
# 4. Heatmap Advantages
cat("\nHeatmap Advantages:\n")
cat("1. Easy to compare student performance.\n")
cat("2. Quickly identifies high and low scores.\n")
cat("3. Helps detect patterns and trends.\n")
cat("4. Useful for decision making and analysis.\n")
cat("5. Provides an attractive visual representation.\n")
55:58
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.2.3 · ~/
> # 4. Heatmap Advantages
> cat("\nHeatmap Advantages:\n")
Heatmap Advantages:
> cat("1. Easy to compare student performance.\n")
1. Easy to compare student performance.
> cat("2. Quickly identifies high and low scores.\n")
2. Quickly identifies high and low scores.
> cat("3. Helps detect patterns and trends.\n")
3. Helps detect patterns and trends.
> cat("4. Useful for decision making and analysis.\n")
4. Useful for decision making and analysis.
> cat("5. Provides an attractive visual representation.\n")
5. Provides an attractive visual representation.
>
Environment
History
Connections
Tutorial
Import Dataset
151 MiB
List
Global Environment
steps_20_30 num [1:10] 9500 10200 9800 11000 10500 9700 10100 ...
steps_31_45 num [1:10] 8200 8600 8400 9000 8900 8500 8700 9100...
steps_46_60 num [1:10] 6500 6700 6900 7200 7000 6800 7100 7300...
student_avg Named num [1:10] 88 74.2 81.2 70 92 63 84 73.6 94....
subject_avg Named num [1:5] 79.1 79.7 78 83.9 79.2
support_students "S6"
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Student Performance Heatmap
88.00
84.00
81.00
S1
76.00
72.00
69.00
80.00
74.00
S2
81.00
78.00
83.00
85.00
79.00
S3
65.00
70.00
68.00
72.00
75.00
S4
S5
58.00
62.00
60.00
65.00
70.00
S6
84.00
86.00
82.00
88.00
80.00
S7
71.00
75.00
73.00
77.00
72.00
S8
S9
79.00
81.00
77.00
83.00
78.00
S10
90
80
70
60
Mathematics
Physics
Chemistry
Programming
English
Activate Windows
Go to Settings to activate Windows.


File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
ro.R*
Source on Save
Run
Source
# 4. Highest Median Group
# ----------------------------
medians <- c(
median(steps_20_30),
median(steps_31_45),
median(steps_46_60)
)
groups <- c("20-30","31-45","46-60")
cat("Median Values:\n")
print(medians)
cat("\nHighest Median Group:",
groups[which.max(medians)], "\n")
74:31
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.2.3 · ~/
+ groups[which.max(medians)], "\n")
Highest Median Group: 20-30
> # ----------------------------
> # 5. Best Visualization
> # ----------------------------
> cat("\nBest Visualization:\n")
Best Visualization:
> cat("Boxplot is the best visualization because it clearly compares the median, spread, and outliers among the three age groups.\n")
Boxplot is the best visualization because it clearly compares the median, spread, and outliers among the three age groups.
Environment
History
Connections
Tutorial
Import Dataset
178 MiB
List
R
Global Environment
steps_20_30 num [1:10] 9500 10200 9800 11000 10500 9700 10100 …
steps_31_45 num [1:10] 8200 8600 8400 9000 8900 8500 8700 9100…
steps_46_60 num [1:10] 6500 6700 6900 7200 7000 6800 7100 7300…
student_avg Named num [1:10] 88 74.2 81.2 70 92 63 84 73.6 94.…
subject_avg Named num [1:5] 79.1 79.7 78 83.9 79.2
support_students "S6"
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Daily Steps by Age Group
Daily Steps
7000
9000
11000
20-30
31-45
46-60
Activate Windows
Go to Settings to activate Windows.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Dhv Act 1.R
Dhv act-2.R*
Dhv Act-3.R
Dhv act 4.R*
Dhv act 5.R
Source on Save
Run
Source
# Average Consumption
avg <- colMeans(electricity)
print(avg)
cat("Month with Highest Average Consumption:", names(which.max(avg)), "\n")
# Standard Deviation
spread <- apply(electricity,2,sd)
print(spread)
cat("Month with Widest Spread:", names(which.max(spread)), "\n")
Console
Terminal
Background Jobs
R 4.2.3
legend=colnames(electricity),
col=1:5,
lty=1,
pch=16)
Jan Feb Mar Apr May
230.8 239.9 256.5 281.2 300.7
Month with Highest Average Consumption: May
Jan Feb Mar Apr May
9.223159 6.279597 5.380004 5.865151 5.538752
Month with Widest Spread: Jan
Environment History Connections Tutorial
Import Dataset
Global Environment
Files Plots Packages Help Viewer Presentation
Zoom
Export
Publish
Monthly Electricity Consumption
Jan
Feb
Mar
Apr
May
Electricity Consumption (kWh)
House Number


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
main="Employee Salary Analysis",
xlab="Departments",
ylab="Salary (Lakhs Per Annum)")
# Standard Deviation
variation <- apply(salary, 2, sd)
print(variation)
cat("Department with Largest Variation:", names(which.max(variation)), "\n")
# Median
med <- apply(salary, 2, median)
print(med)
cat("Department with Highest Median Salary:", names(which.max(med)), "\n")
Console Terminal Background Jobs
R 4.2.3
+ col=c("lightblue","lightgreen","pink","yellow"),
+ main="Employee Salary Analysis",
+ xlab="Departments",
+ ylab="Salary (Lakhs Per Annum)")
> # Standard Deviation
> variation <- apply(salary, 2, sd)
> print(variation)
HR IT Finance Marketing
0.2581989 0.7554248 0.4552167 0.3011091
> cat("Department with Largest Variation:", names(which.max(variation)), "\n")
Department with Largest Variation: IT
> # Median
> med <- apply(salary, 2, median)
> print(med)
HR IT Finance Marketing
4.50 7.35 6.25 5.45
> cat("Department with Highest Median Salary:", names(which.max(med)), "\n")
Department with Highest Median Salary: IT
Environment History Connections Tutorial
Files Plots Packages Help Viewer Presentation
Employee Salary Analysis
Salary (Lakhs Per Annum)
4
5
6
7
8
HR
IT
Finance
Marketing
Departments



RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
act 4-5.R*
Source on Save
Run
Source
# Monthly Income
income <- c(18000,22000,24000,26000,28000,
30000,34000,42000,85000,200000)
Histogram
hist(income,
main = "Histogram of Monthly Income",
xlab = "Income",
col = "lightgreen")
# Q-Q Plot
qqnorm(income,
main = "Normal Q-Q Plot of Income")
qqline(income, col = "red", lwd = 2)
14:37
(Top Level)
R Script
R 4.2.3 · ~/
+ main = "Normal Q-Q Plot of Waiting Time")
> qqline(waiting, col = "red", lwd = 2)
> # Monthly Income
> income <- c(18000,22000,24000,26000,28000,
+ 30000,34000,42000,85000,200000)
>
> # Histogram
> hist(income,
+ main = "Histogram of Monthly Income",
+ xlab = "Income",
+ col = "lightgreen")
>
> # Q-Q Plot
> qqnorm(income,
+ main = "Normal Q-Q Plot of Income")
> qqline(income, col = "red", lwd = 2)
>
Environment History Connections Tutorial
Files Plots Packages Help Viewer Presentation
Zoom
Export
Publish
Normal Q-Q Plot of Income
Sample Quantiles
200000
150000
100000
50000
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
Theoretical Quantiles